1 Initialize document weights

computed once before decoding



Query *x*

How many eggs does this bird lay?

Retriever
selects set Z

knowledge base

Z

K singleton inputs

(x, z_1)

(x, z_2)

$\vdots$

(x, z_K)

VLM
shared Θ
K parallel passes

Last-token embeddings

$e_\Theta(x,z_1)$

$e_\Theta(x,z_2)$

$\vdots$

$e_\Theta(x,z_K)$

MLP
$s(\cdot)$
softmax

Document prior

$$m_{1,k} = P_\Theta(z_k \mid x,Z) = \frac{exp\ s(e_\Theta(x,z_k))}{\Sigma_{k'=1}^{K}\ exp\ s(e_\Theta(x,z_{k'}))}$$

Eq. 5

2 Decode token j and update the posterior

repeat until EOS

K parallel VLM branches

$(x,z_1,y_{<j})$

$(x,z_2,y_{<j})$

$\vdots$

$(x,z_K,y_{<j})$

Per-document distributions

$$p_{j,k}(\cdot) = P_\Theta(\cdot \mid y_{<j},x,z_k)$$

Posterior-weighted mixture

$$P(y_j \mid y_{<j},x,Z) = \Sigma_{k=1}^{K} P_\Theta(y_j \mid y_{<j},x,z_k)\ P(z_k \mid y_{<j},x,Z)$$

Eq. 2

Decode
greedy / sampling

y_j

Current document posterior

$$m_{j,k} = P(z_k \mid y_{<j},x,Z)$$

Eq. 3

Recursive Bayes update for the next step

$$m_{j+1,k} = \frac{P_\Theta(y_j \mid y_{<j},z_k,x)\ m_{j,k}}{\Sigma_{k'=1}^{K} P_\Theta(y_j \mid y_{<j},z_{k'},x)\ m_{j,k'}}$$

Eq. 4

use updated posterior at step j + 1